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GATE ECE · ANALOG CIRCUITS · CHAPTER-WISE PYQS

Diodes

GATE ECE Analog Circuits · Previous Year Questions with Answer Key

22	18	1998 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 1	12 questions
Marks 2	10 questions
Answer key	at the end

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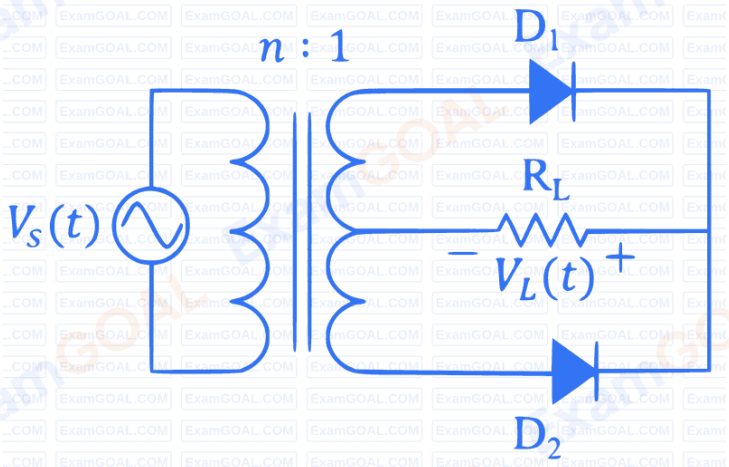
1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.
3. **Understand**
   Scan the QR code beside any question for its full solution.

Marks 1 12 questions

Q1. GATE ECE 2024

Open solution

In the circuit shown, the n : 1 step-down transformer and the diodes are ideal. The diodes have no voltage drop in forward biased condition. If the input voltage (in Volts) is $V_s(t) = 10 \sin \omega t$ and the average value of load voltage $V_L(t)$ (in Volts) is $\frac{2.5}{\pi}$, the value of n is ____.

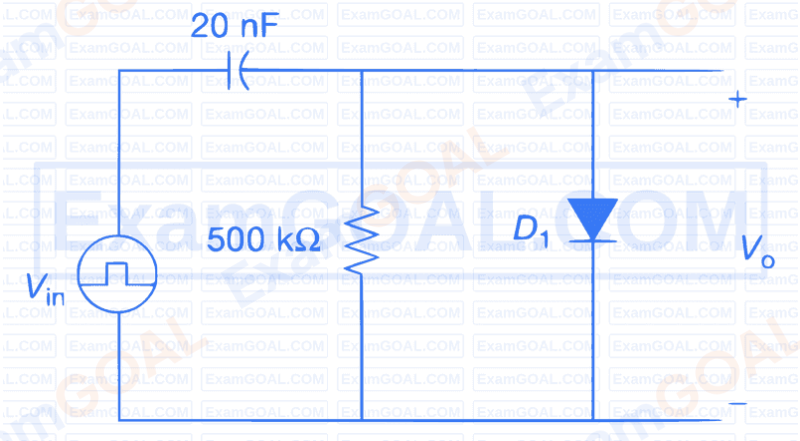


(A) 4 (B) 8 (C) 12 (D) 16

Q2. GATE ECE 2021

Open solution

An asymmetrical periodic pulse train v_{in} of 10 V amplitude with on-time $T_{\text{ON}} = 1\mu\,\text{s}$ is applied to the circuit shown in the figure. The diode D_1 is ideal.



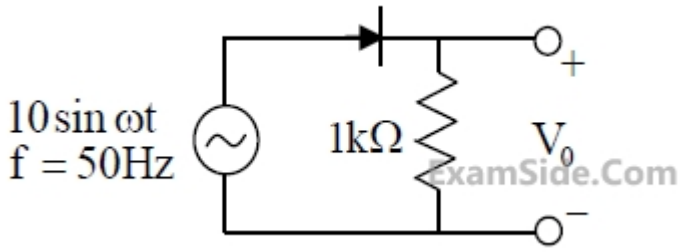
The difference between the maximum voltage and minimum voltage of the output waveform v_0 (in integer) is ____ V.

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Q3. GATE ECE 2017 Set 2

The output V_0 of the diode circuit shown in the figure is connected to an averaging DC voltmeter. The reading on the DC voltmeter in Volts, neglecting the voltage drop across the diode, is _____.

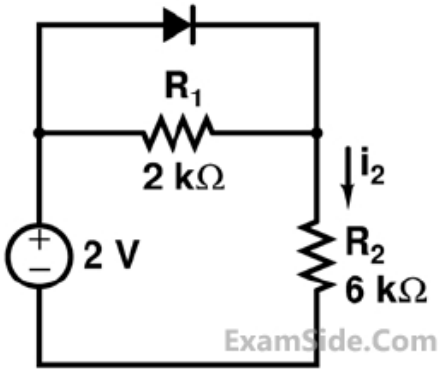
Open solution



Q4. GATE ECE 2016 Set 2

Assume that the diode in the figure has on V_{on} = 0.7V, but is otherwise ideal.

Open solution



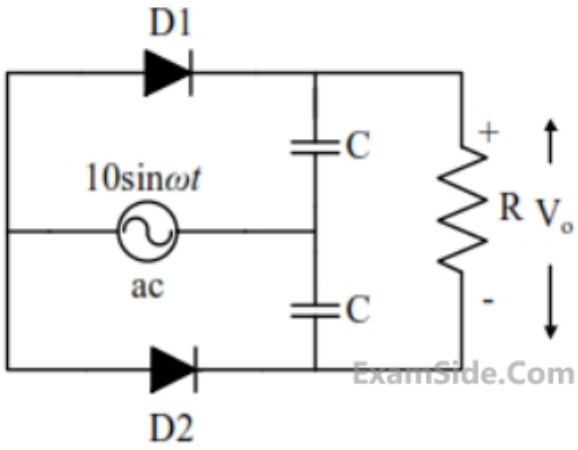
The magnitude of the current i_2 (in mA) is equal to _____.

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Q5. GATE ECE 2016 Set 3

The diodes D1 and D2 in the figure are ideal and the capacitors are identical. The product RC is very large compared to the time period of the ac voltage. Assuming that the diodes do not breakdown in the reverse bias, the output voltage V_0 (in volt) at the steady state is_____.

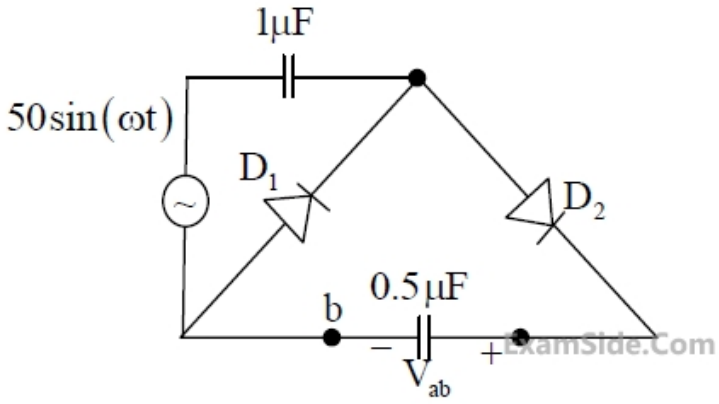
Open solution



Q6. GATE ECE 2015 Set 3

In the circuit shown, assume that diodes D_1 and D_2 are ideal. In the steady-state condition the average voltage V_{ab} (in Volts) across the 0.5 μF capacitor is _____.

Open solution

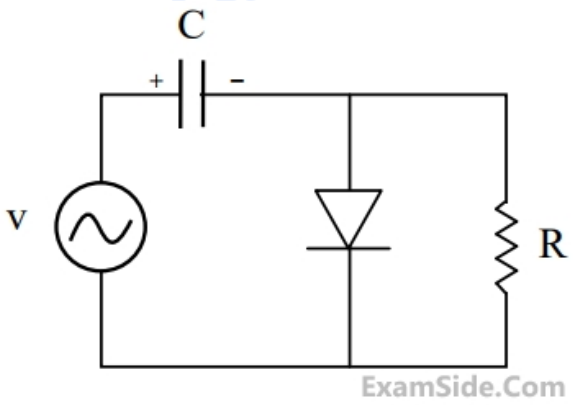


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Q7. GATE ECE 2015 Set 2

Open solution

If the circuit shown has to function as a clamping circuit, which one of the following conditions should be satisfied for sinusoidal signal of period T?



(A) RC << T (B) RC = 0.35 T (C) RC ≈ T (D) RC >> T

Q8. GATE ECE 2014 Set 4

Open solution

For a given sample-and-hold circuit, if the value of the hold capacitor is increased, then

(A) Droop rate decreases and acquisition time decreases

(B) Droop rate decreases and acquisition time increases

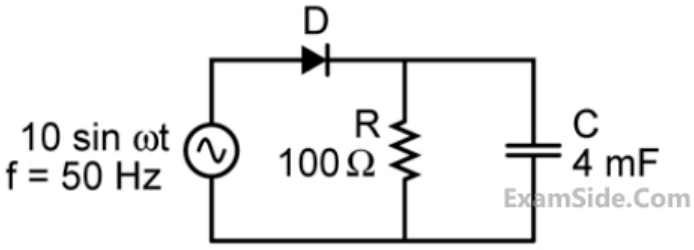
(C) Droop rate increases and acquisition time decreases

(D) Droop rate increases and acquisition time increases

Q9. GATE ECE 2014 Set 3

Open solution

The figure shows a half-wave rectifier. The diode D is ideal. The average steady-state current (in Amperes) through the diode is approximately _____.

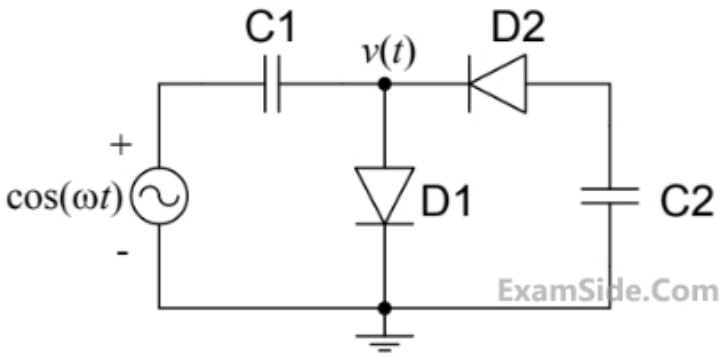


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Q10. GATE ECE 2012

The diodes and capacitors in the circuit shown are ideal. The voltage v(t) across the diode D1 is

Open solution

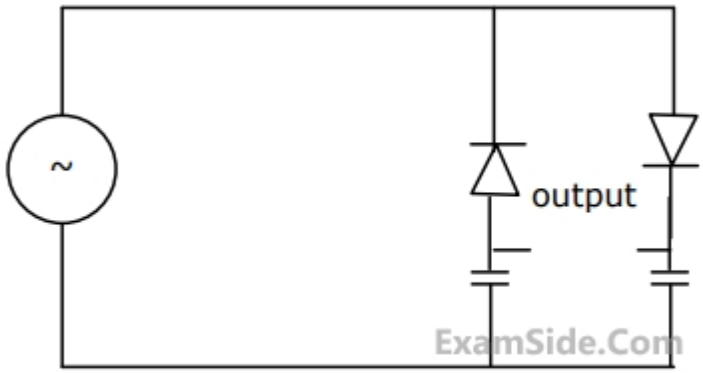


(A) cos (ωt) – 1 (B) sin (ωt) (C) 1 - cos (ωt) (D) 1 - sin (ωt)

Q11. GATE ECE 2003

The circuit shown in figure is best described as a

Open solution



(A) bridge rectifier
(B) ring modulator
(C) frequency discriminator
(D) voltage doubler

Q12. GATE ECE 1998

For full wave rectification, a four diode bridge rectifier is claimed to have the following advantages over a two diode circuit:
(1) less expensive transformer
(2) smaller size transformer, and
(3) suitability for higher voltage application.
Of these,

Open solution

(A) only (1) and (2) are true
(B) only (1) and (3) are true
(C) only (2) and (3) are true
(D) (1), (2) as well as (3) are true

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Marks 2 10 questions

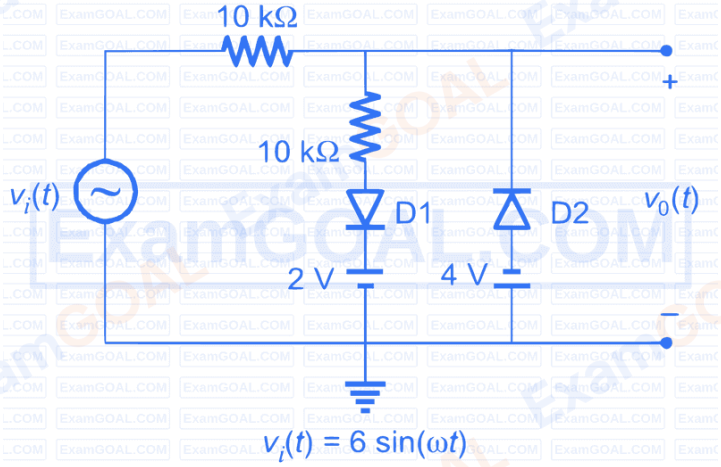
Q13. GATE ECE 2026

Open solution

Consider the ideal diodes D1 and D2 as shown in the Figure with cut-in voltage $V_\gamma = 0$ Volt and $v_i(t)$ is in Volt.

The maximum voltage (Volt) of the output $v_o(t)$ is ____ .

(rounded off to two decimal places)



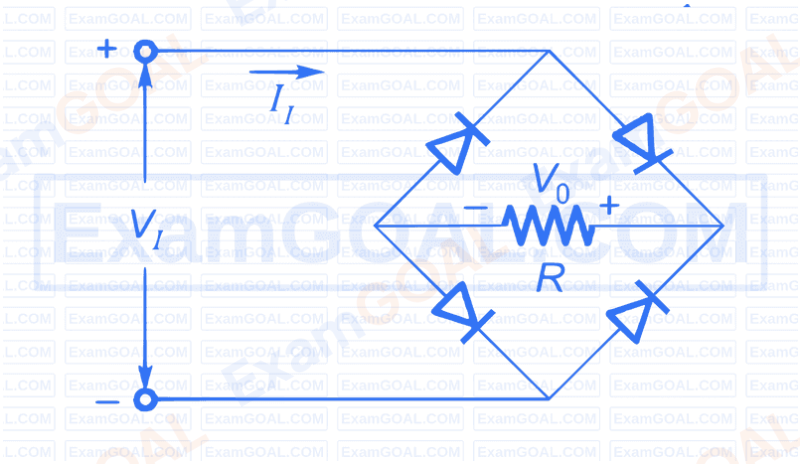
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Q14. GATE ECE 2025

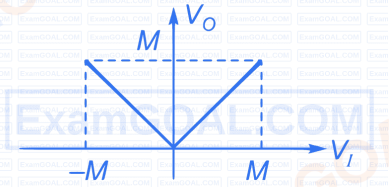
Open solution

All the diodes in the circuit given below are ideal.

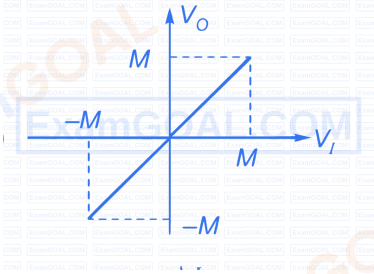
Which of the following plots is/are correct when V_I (in Volts) is swept from $-M$ to M ?



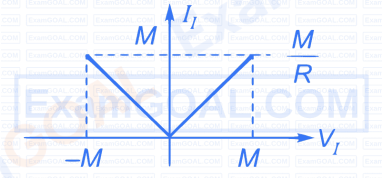
(A)



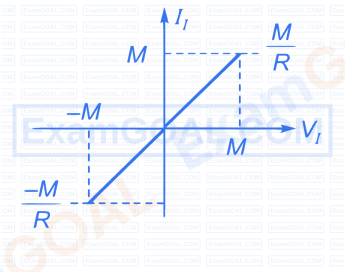
(B)



(C)



(D)



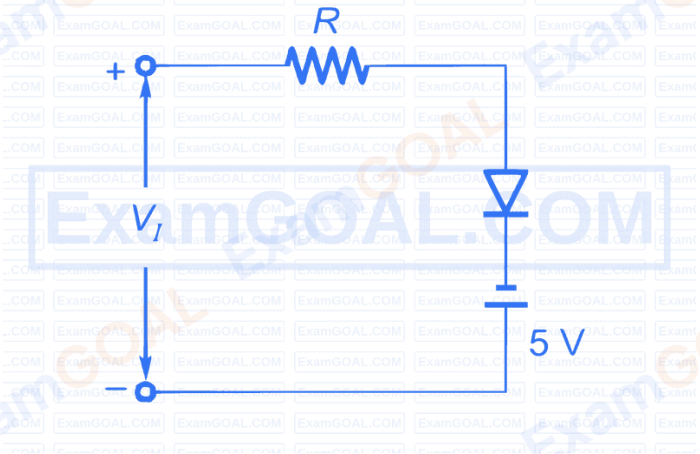
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Q15. GATE ECE 2025

The diode in the circuit shown below is ideal. The input voltage (in Volts) is given by $V_1 = 10 \sin 100\pi t$, where time t is in seconds.

The time duration (in ms, rounded off to two decimal places) for which the diode is forward biased during one period of the input is _____.

Open solution

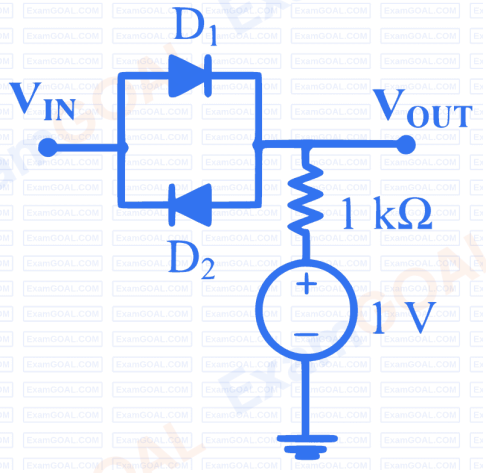


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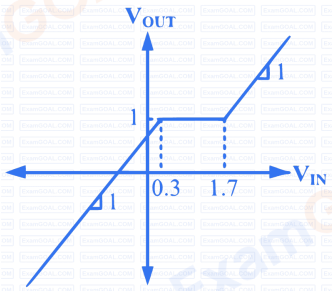
Q16. GATE ECE 2023

Open solution

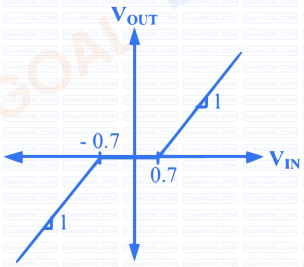
In the circuit shown below, D_1 and D_2 are silicon diodes with cut-in voltage of 0.7 V. V_{IN} and V_{OUT} are input and output voltages in volts. The transfer characteristic is



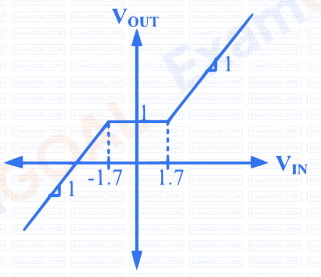
(A)



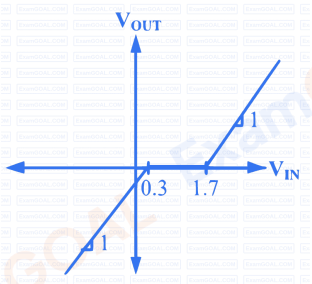
(B)



(C)



(D)

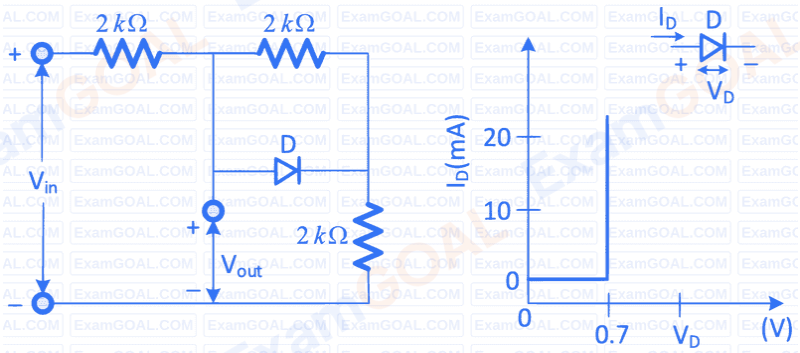


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Q17. GATE ECE 2022

Open solution

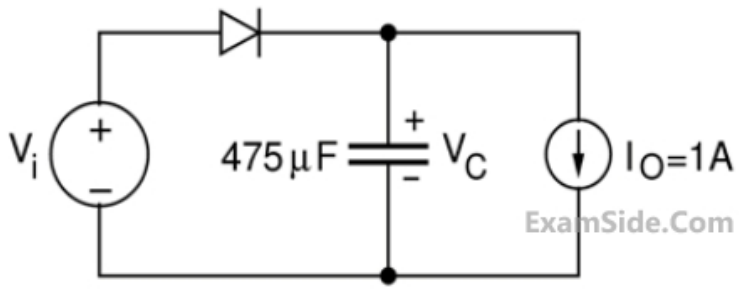
A circuit and the characteristics of the diode (D) in it are shown. The ratio of the minimum to the maximum small signal voltage gain $\frac{\partial V_{out}}{\partial V_{in}}$ is _____ (rounded off to two decimal places).

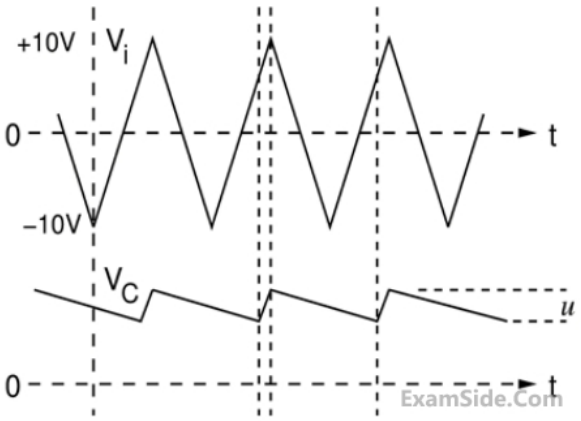


Q18. GATE ECE 2016 Set 2

Open solution

The figure shows a half-wave rectifier with a 475 μF filter capacitor. The load draws a constant current I_0= 1 A from the rectifier. The figure also shows the input voltage V_i , the output voltage V_C and the peak-to-peak voltage ripple on V_C.The input voltage V_i is a triangle-wave with an amplitude of 10 V and a period of 1 ms.





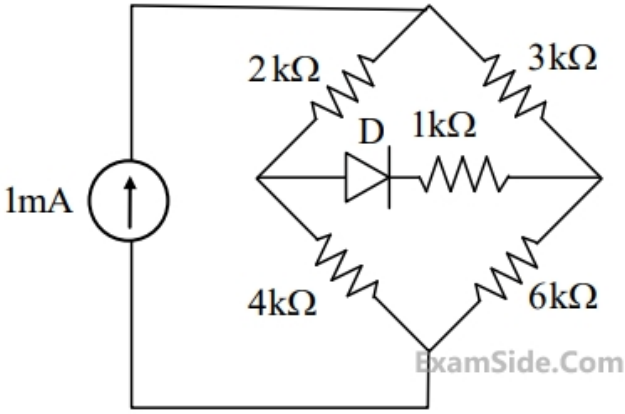
The value of the ripple u (in volts) is _____.

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Q19. GATE ECE 2015 Set 2

Open solution

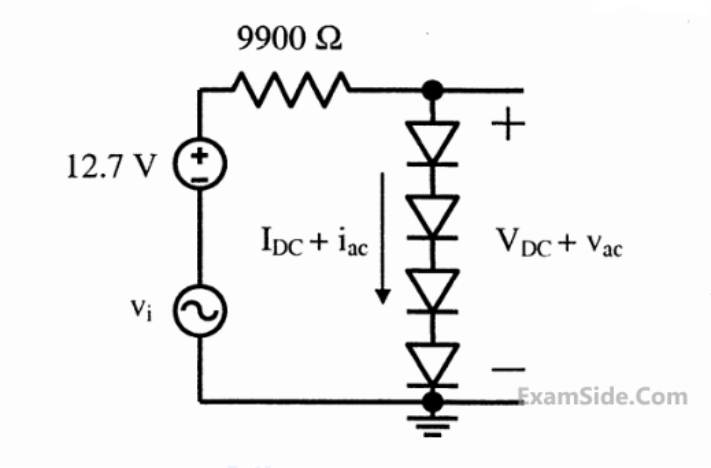
The diode in the circuit given below has V_{ON} = 0.7V but is ideal otherwise. The current (in mA) in the 4kΩ resistor is _____.



Q20. GATE ECE 2011

Open solution

In the circuit shown below, assume that the voltage drop across a forward biased diode is 0.7V. The thermal voltage V_t = kT/q = 25mV. The small signal input $v_i = V_P \cos(\omega t)$ where V_P = 100mV.



The bias current I_{DC} through the diodes is

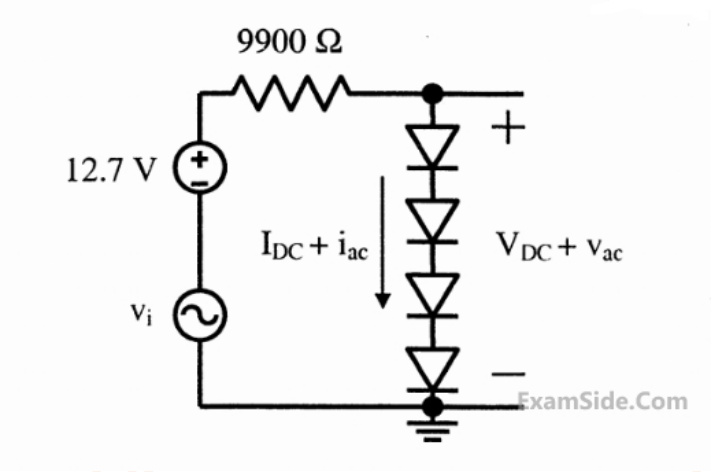
(A) 1 mA (B) 1.28 mA (C) 1.5 mA (D) 2 mA

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Q21. GATE ECE 2011

In the circuit shown below, assume that the voltage drop across a forward biased diode is 0.7V. The thermal voltage $V_t = kT/q = 25mV$. The small signal input $v_i = V_P \cos(\omega t)$ where $V_P = 100mV$.

Open solution



The ac output voltage v_{ac} is

(A) $0.25\cos(\omega t)$mV (B) $1\cos(\omega t)$mV (C) $2\cos(\omega t)$mV (D) $22\cos(\omega t)$mV

Q22. GATE ECE 1999

A dc power supply has a no-load voltage of 30V, and a full-load voltage of 25 V at a full-load current of 1A. Its output resistance and load regulation, respectively are

(A) 5Ω and 20% (B) 25Ω and 20% (C) 5Ω and 16.7% (D) 25Ω and 16.7%

Open solution

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Answer Key

Marks 1

1. **A**	2. **10 to 10**	3. **3.18**	4. **0.25**	5. **0**
6. **100**	7. **D**	8. **B**	9. **0.1**	10. **A**
11. **D**	12. **D**			

Marks 2

13. **(4) (3.70 to 4.30)**	14. **A, D**	15. **13.32 to 13.34**	16. **A**	17. **0.75**
18. **2.1**	19. **0.6**	20. **A**	21. **B**	22. **B**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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